

JORDI CORBILLA

SENIOR/LEAD SOFTWARE ENGINEER / LEAD ARCHITECT

A multi-skilled Senior Software Engineer with good all-round ability to lead multiple projects and meet deadlines whilst at the same time comprehending complex and interdependent business processes and products. Professional with strong computer science fundamentals and 19+ years of Software engineering experience with 12 years as a Lead Engineer with a very stable career history across multiple industries and sectors including finance, fund management, and industrial automation. Very capable with an ability to identify and then deal with a client's needs by translating them into appropriate technical solutions. Proven mentor and trainer, skilled at communicating with all organisational levels and cross-functional teams to develop a shared vision and foster a culture of excellence. Strong leadership abilities, motivating to embrace new challenges and work collaboratively together to achieve goals. Inspires others to excel by gaining buy-in, communicating openly and building mutual respect.

EXPERTISE

- | | | |
|----------------------------|----------------------------|-----------------------------|
| • Management & Leadership | • TDD, BDD, DevOps & SOLID | • Continuous Improvement |
| • Talent Development | • Project Management | • Change Management |
| • Order Management System | • Mentoring & Coaching | • Problem Solving |
| • Financial Tech Solutions | • Solutions Architecture | • UI/UX design |
| • Hedge Funds | • Microservices & Cloud | • Negotiating & influencing |
| • Optimization | • Machine Learning | • Code Reviews |

EXPERIENCE

Balyasny Asset Management – London (UK)

Senior Software Engineer, Equity Trading & Compliance Technology

09/2020 - present

- Design, build and extend BAM's trading and compliance technology platforms using cloud services, containers, REST API and React.
- Work closely with trading, compliance and technology teams to deliver cutting-edge innovative software solutions to businesses and to uphold a diverse set of industry regulations and requirements and help better equip BAM to enter new markets with increasingly complicated regulatory needs.
- MiFID II Trade Regulatory Reporting for T0 and T1.
- Market Data / Pricing APIs via BPIPE.
- MAS OTC Regulatory Reporting.
- Designing, Scaling, optimizing and refactoring large-scale systems, and with diverse data locality.
- Pre-trade surveillance and research process automation, trade execution rules and exposures automation, ownership, short reporting, post-trade allocation, and regulatory filings' automation.
- Market abuse automation, restrictions and exclusions.

Notable Achievements

- Automated several complex manual systems to reduce the TCO for the Compliance & Trading Teams.
- MAS OTC Reporting for Equities, FX and Commodities.
- Introduced SpecFlow testing as part of the micro-services approach.
- Cloud adoption using AWS, Harbor, Rancher, Helm, python, .NET 6, React, Docker and Kubernetes.
- Mentoring internship fellows as part of the [BAM ATLAS program](#).

VR Capital Group – London (UK)

Director of Application Development

02/2019 – 09/2020

- Full stack 100% hands-on Director, drove the end-to-end enterprise solutions and the delivery of end-to-end architectures and provided full digital transformations of new products and services. Owned the deliveries, quality, and timelines.
- Approached projects with a forward-thinking mindset and with disrupting technologies.
- Improved route-to-live processes using DevOps, testing automation and quality control.
- Directed and coached distributed agile teams, making sure there is accountable incremental transformation daily to increase the value we deliver through technology to our business stakeholders.
- Performance optimisation & built robust intelligent solutions to problems.
- Innovated whilst bringing in new ideas.
- Took full ownership of the end-to-end delivery/deployment, technical roadmaps, choice of technologies and implementation of best practices and KPIs.
- Architected and developed critical-mission finance/trading systems using C#, Python, ASP.NET core, SQL Server, TDD, Docker, Kubernetes, responsive websites (HTML5), React and Blazor.

Notable Achievements

- Led the OMS system project on a 4 year-plan to add 20 different asset types (FX, Repo, FI, Equities, Futures, Swaps, CDS, OTC, etc) including P&L interaction, Risk and Compliance capabilities, supporting thousands of trades per day and web/mobile availability.
- Led all internal financial products and total reward systems (bonus scheme, salaries, trading limits).
- Led a team of developers and testers based in Ukraine with a \$1m/year budget.
- Led the infrastructure migration to achieve high availability.
- Led the Agile transformation and DevOps culture adoption.

Lead Software Engineer / Solutions Architect

02/2016 - 02/2019

- Architected and developed in-house investment, trading, analytics systems using ASP.NET, WPF, Rx, PHP, Scrum, C#, SQL Server, TDD, MVC, JavaScript, Python, CSS, jQuery, SignalR, Knockoutjs, Angular, Reactjs, XML, API, Web Services, HTML, DHTML, Rx, Design Patterns, and REST. (Full Stack).
- Provided support to trading desks across multiple locations (NY, Moscow, Dubai and London).
- Led the design and specification of all system components considering functionality, data, security, integration, infrastructure and performance.
- Multiple project handling whilst assessing other Developer's workload.
- Contributed to the platform redesign and architectural choices to shape the future of technology for the group.
- Collaborated with remote development teams (NY, Ukraine), to build a true global team.
- Helped to establish a long-lasting and repeatable software development standard across the team using Agile methodologies.
- Participated in code reviews across all areas of the team, to improve the quality of the delivered solutions.
- Contributed to the overall security and vulnerabilities of the applications. (OWASP).
- Contributed to the establishment of a culture of TDD, BDD and CD.
- Worked across green-field projects to support new market opportunities, improve process efficiencies, reduce operational risk, improve platform scalability and regulatory changes.
- Worked collaboratively with trading desks worldwide to deliver strategic end objectives.
- MIFID II Trade Regulatory Reporting, RHUB and FIX Messaging.
- Implement Compliance solutions to store all messaging/email and phone conversations as part of the internal discovery.
- Supported legacy ASP.NET and PHP applications.
- Delivered high quality IT Service within the company.
- Worked with experienced .NET C# developers to help design, code and test a wide range of key business software applications that are utilised throughout Front, Middle and Back Office business areas.

Notable Achievements

- Led the internal change to achieve a culture of continuous delivery & continuous innovation.
- Worked closely with Traders & PMs to automate several aspects of their work (FX roll-overs, model automation, ownership alerts, shorts filing, etc).
- Created the internal architecture that allowed the onboarding of many teams and the early adoption of DevOps.
- Obtained a directorship role at VR Capital due to subject matter expertise and success during the previous two years.
- Delivered 70% reduction in time across departments, increased efficiency and productivity by reducing human error in data entry.

Broadridge Financial Solutions Ltd – London (UK)

Lead Technical Architect

03/2011 - 02/2016

- Enhanced current solution [PROactive™ Confirmation Matching](#).
- Executed and developed end-to-end software solutions in agile environment including: requirement analysis, architecture, design, coding, testing, deployment and support.
- Business process knowledge including trade matching, HFT (High frequency trading), settlement and SWIFT.
- Developed of scalable systems with architectural requirements for high availability, high throughput and low response time.
- Object-Oriented Analysis and Design (OOA/OOD) with appropriate use of design patterns.
- Handled of huge data and processing large amounts of information providing high response systems under SQL Server 2008 R2/2012/14. (Analysis and performance).
- Unit testing with DUnit, code coverage and continuous integration (TeamCity).
- Drove continual improvement to system architecture by refactoring old legacy code.
- Designed systems or components on large projects (design, configuration, coding, testing, implementation and documentation).
- Modelled mathematic algorithms to display performance rates for multi concurrent entities.
- Expert with source control management using TFS, Subversion and GitHub.
- Architecture domain skills (Data, Application, and Infrastructure) using a good understanding of architectural concepts, methodologies and mapping approaches.
- Project management of new products making key technical decisions and guiding and mentoring the team around technical challenges.
- Experience of working with outsourced or offshore partners to deliver software projects.
- Collaborated with other parts of the technology team, including product, QA, etc. to deliver solid product.
- Solid experience (+5 years) in testing methodologies, test-driven design (TDD) and the creation of effective and reliable test cases.
- Advocated the use of the Domain Driven Design (DDD), Inversion of Control, good Object-oriented design practices, MVC, Single Responsibility Principle, Gang of Four Design Patterns, Open/Closed Principle, Simplicity of Design, UML, TDD, Code Coverage, BDD and other similar practices that help create solid, maintainable code.
- Conduct code review providing feedback.

- Mentoring/Training people in India to increase their technical ability.
- Travelled to India several times to increase the technical skills of my team.
- Reviewed technical requirements provided by the business for key projects, in order to take concepts from design to delivery. Provided solid software/component level designs supporting technical requirements.
- Brought teams to a higher level of performance.
- Highly efficient in Agile processes. Agile mindset, being able to establish continuous deployment processes. Breaking down concepts into user stories. Gather data and provide Agile metrics: FTR, sprint burn down, etc. Scrum dashboards, daily/weekly scrum calls with the team. Retrospectives, plan releases, etc.
- Expertise with Continuous Integration (CI), Continuous Deployment and Continuous Delivery and used in several projects across the company.
- Developed automation testing suite for cross-browser testing (Chrome, Firefox, IE and Safari) using Ruby, Watir and BrowserStack.
- Configured and set up Build Agents infrastructure to perform Continuous deployment. VMWare machines with WebDeploy and TeamCity.

Notable Achievements

- Architected and developed new solution: [PROactive™ Matching High Frequency Trading](#) to cope with algorithmic trading and HFT (High frequency trading) - 1 million trades/day. The market leading confirmation matching solution (Treasury Management System) that automates the processing of trade confirmations with unbeatable match rates and exception workflow from the earliest point in the pre-settlement cycle. The new state-of-the-art HFT solution has been in development under my supervision using my architectural solutions over 24 months and implemented in Bank of America New York. This solution works under Windows platform using concurrent modules. This solution has been driven through agile programming methods. Delphi and .Net technologies used over SQLServer.
- Architected and developed the solution [PROactive™ eMatching](#) for the last 24 months to provide a web portal for internal banking and corporates. This solution uses ASP.NET, MVC, JavaScript, CSS, jQuery, SQLServer and other .NET technologies. The project has been under my supervision with a team of 8 developers: 4 in the UK and other 4 in India. The project is live at Commerz Bank, handling thousands of users and thousands of requests per second.

[AEtech - Girona \(Spain\)](#)

Senior Software Engineer

08/2007 – 03/2011

- Handled project architecture, conceptual & functional design, development & deployment while keeping in mind the "big picture", making sure that "small details" are properly addressed in the implementation.
- Full DB integration, design and performance optimization with SQL Server 2000 - 2005 backend.
- Proposed and managed implementation of proper DB versioning and deployment procedures.
- Well proficient in organizing and prioritizing task to accomplish desired results in most efficient way.
- Designed and implemented necessary deployment processes related to applications managed. Developed necessary automated scripts and created and maintained required support documentation.
- Experience with third-party VCL Components (DevExpress), knowledge of Win32 API, TCP/IP Protocols and socket programming.
- Ability to design and develop in OOAD - Object Orientated Analysis / Design (UML, Design Patterns).
- Database experience in SQL. UML (Sequence Diagrams, Class Diagrams, Entity Relationship Diagrams and Deployment Diagrams).

Notable Achievements

- Designed and managed implementation of Company's front-end clients (Win32 with Delphi 2007-XE) and core Application Server modules - infrastructure, data feeds, multithreading data processing and analysis, data manipulation and aggregation.
- Developed internal framework scheduling system solution (AEmes Scheduling) in collaboration with UPC (Polytechnic University of Catalonia) To provide necessary functionality for increasing the productivity in batch factories. The code is exclusively Delphi with widespread use of VCL for the GUI front-end.
- Developed a timing sequencing algorithm for cluster tasks base on the following paper [[Design and development of a timing sequencing algorithm for cluster tasks](#)].
- Developed an internal Open-source framework ([Zaluum](#)) For industrial automation based on Java in collaboration with UPC. Setting up the multithreading system on a RTOS (Real time OS) with Ubuntu Server and JVM RT (Real time Java VM).
- Developed [AEmes](#) (Manufacturing Execution System) ERP product server side / client side. Improved AI algorithms to achieve high performance with low latency.
- Developed a library in C# to automate the drawing of schemes in Autocad 2009 using COM and Delphi.

Senior Automation Control Engineer

06/2003 – 08/2007

- Strong technical knowledge in PLC architecture and Programming, HMI/SCADA Programming, DCS Systems.
- Drive commissioning, soft starters commissioning and PID design.
- Technical documentation, monitored work for large customers, support and technical service.
- Provided technical support for production operations including troubleshooting equipment and process problems.
- Actively participated in all the phases of assigned projects including design, fabrication, off-line integration and testing, installation, production start-up support, training of plant personnel and transfer to operations.
- Expertise in a full array of automation technologies including PLC and HMI programming, panel design and system integrations.
- Control software and Electrical hardware troubleshooting.
- Field support for installing/checkout of design and make suggestions for improvements.

- Familiarity with Minitab and statistical process control.
- Hands-on knowledge of electrical installation, self-starter, independent and quick learner.
- Data retrieval, plotting and analysis using Microsoft Excel.
- Experienced with MES application development.
- Interested in outstanding engineering feats, industrial automation, robotics, manufacturing and process industries.
- Provided 24x7 on call support.

Notable Achievements

- Created simulation models for production improvements using Matlab.
- Developed new or improved existing product designs to address critical quality requirements.
- Developed methods, measurements and processes to improve capabilities related to manufacturing.
- Improved hundreds of manufacturing processes and products.

EDUCATION & PERSONAL DEVELOPMENT

2014 - 2019 **MSc in Computer Engineering**. Open University of Catalonia (Spain).

Area of emphasis: Advanced AI, Machine Learning, Distributed Systems.

Master's thesis: [Ocular disease intelligent recognition through Deep learning architectures](#).

2006 - 2014 **BSc (Hons) Computer Engineering**. Open University of Catalonia (Spain).

Area of emphasis: Grammars, Compilers, Knowledge representation, Data mining, computerised learning and AI.

Bachelor's thesis: [Distributed decision-making system](#).

1999 - 2003 **BSc (Hons) Industrial Electronics Engineering**. University of Girona (Spain).

Area of emphasis: Electronics, Robotics and Industrial Automation.

Bachelor's thesis: [Monitoring of a remote laboratory via Data Socket](#).

Latest certification and training:

- 2022 Python for Data Science and Machine Learning, Udemy.
- 2022 Learn to build an e-commerce store with .net 6, React & Redux, Udemy.
- 2022 Introduction to Unity for 2D Video Games (Ongoing). Domestika.
- 2021 AWS Solutions Architect Certification (Ongoing), Udemy.
- 2021 Crypto Trading with QuantConnect, Udemy.
- 2021 The Complete Cryptocurrency Investment course, Udemy.
- 2021 Data Science: Natural Language Processing (NLP) in Python, Udemy.
- 2021 Stock Trading: Invest and Trade Safely with Stock Sector ETF, Udemy.
- 2021 React – The Complete Guide (Hooks, React Router, Redux), Udemy.
- 2020, ASP.NET Core API (.NET 5.0) and Angular 10, Udemy.
- 2020, Algorithmic Trading & Quantitative Analysis Using Python, Udemy.
- 2020, The Complete Foundation Stock Trading Course, Udemy.

TECHNICAL EXPERTISE

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| <ul style="list-style-type: none"> • Data Storage: SQLServer 2012-2019, MySql, Oracle • FIX Systems: QuickFixN, FIXEdge • Languages: C#, Python, Delphi, Java, C++, C, VB6, Ruby, R, Lua, Powershell • Web technologies: ASP.NET Webforms, ASP.NET MVC 5, Web API's (REST, json), jQuery, JavaScript, CSS, Ajax, HTML 5, ASP.NET Core 3.1/5/6, SignalR, Knockoutjs, Reactjs, Angular 10, Blazor, Swagger. • Desktop: Winforms, WFP, VCL, FMX, .NET 4.8, .NET Core 3.1, .NET 5/6 | <ul style="list-style-type: none"> • OMS / EMS Applications: Bloomberg, FlexTrade, Traiana, MS Passport • Middle Office: Geneva and Arcesium • Technology: TCP/IP, COM, DCOM, ActiveX, WCF, Entity Framework, Multi-threading, HPC, AI and ML • CI/CD: TeamCity, Octopus, Prometheus and Grafana • Scheduling: Active Batch • Mobile: Android Studio, PhoneGap • Logging: Serilog, Log4net • Components: TelerikUI, DevExpress • Security: OWASP, SSL, HTTPS, IIS Configuration, Pen Testing | <ul style="list-style-type: none"> • Regulatory Reporting MiFID II/MAS OTC: TradEcho, RHUB, Unavista, SS&C, DTCC. • Source control: TFS, Git, Svn • Messaging: IBM WebSphere, MSMQ, Kafka, RabbitMQ. • Cloud: Azure, AWS (S3, CloudFront). • Agile Applications: JIRA, Confluence, Redmine and TFS • Containers: Docker and Kubernetes • Deep Learning: Tensorflow and Keras • Reporting: SSRS • Testing: SpecFlow, MSTest, Nunit • Data Analysis: Python (NumPy, SciPy and Pandas) |
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ADDITIONAL INFORMATION

- **Nationality:** British
- **Languages:** Spanish (Native), Catalan (Native), English (Fluent), Portuguese (Intermediate)
- **Interests:** Programming and Photography
- **Awards:** [Embarcadero MVP](#)
- **Blog:** [Random thoughts on coding and technology](#)
- **GitHub:** <https://github.com/JordiCorbilla>
- **LinkedIn:** <https://www.linkedin.com/in/jordicollcorbilla/>